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SECTION; BSIT - 2A

IT Automation Incident Ticket Manager Using Linear Data Structure

OBJECTIVE

The objective of this laboratory activity is to develop a Python program that manages incident tickets generated by automation bots. The program demonstrates basic operations of a linear data structure such as adding, displaying, searching, removing, and counting incident tickets.

SCENARIO

The program simulates an IT Automation Incident Ticket Manager. Automation bots can generate different incident tickets when errors occur in processes such as inventory reporting, email notifications, data transfer, invoicing, and backups.

The program allows the user to manage these incident tickets through a menu-driven system.

DATA STRUCTURE USED

The program uses a **Python list**, which is a linear data structure.

Each element of the list represents one incident ticket. Each ticket contains three pieces of information:

- **Incident ID**
- **Bot**
- **Short Description**

The list allows the program to add new tickets, display existing tickets, search for tickets, remove resolved tickets, and count active tickets.

PROGRAM OPERATIONS

1. Add

The `append()` function is used to add a new incident ticket to the list.

```
incident_tickets.append(new_ticket)
```

2. Display

A for loop is used to display all active incident tickets.

3. Search

The program searches the list using the Incident ID.

4. Remove

The `remove()` function removes a resolved incident ticket.

```
incident_tickets.remove(ticket)
```

5. Count

The `len()` function counts the active incident tickets.

```
len(incident_tickets)
```

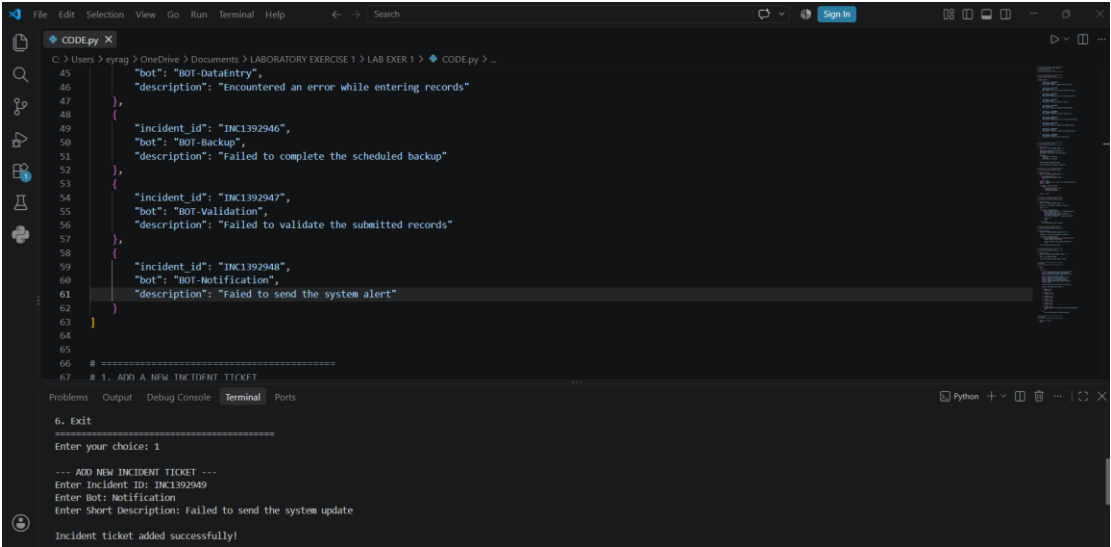
Source Code



CODE.py

Program Output

Adding a Ticket



The screenshot shows a code editor with a Python script and its terminal output. The script defines a list of incidents and a function to add a new incident ticket. The terminal shows the user entering '1' to add a new ticket, followed by prompts for incident ID, bot name, and short description, resulting in a successful addition.

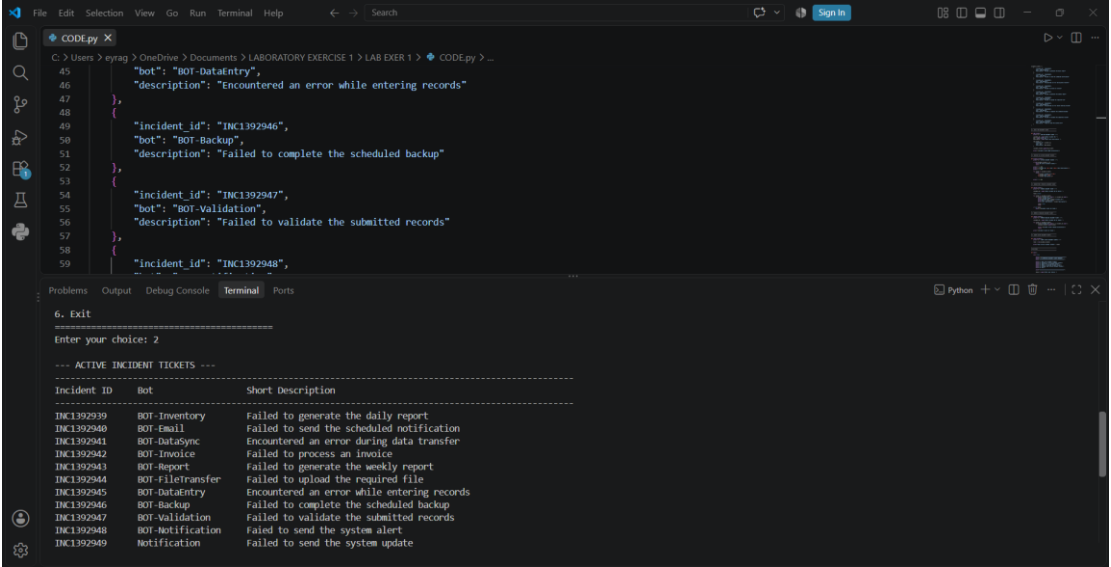
```
45     "bot": "BOT-DataEntry",
46     "description": "Encountered an error while entering records"
47 },
48 ],
49     "incident_id": "INC1392946",
50     "bot": "BOT-Backup",
51     "description": "Failed to complete the scheduled backup"
52 },
53 ],
54     "incident_id": "INC1392947",
55     "bot": "BOT-Validation",
56     "description": "Failed to validate the submitted records"
57 },
58 ],
59     "incident_id": "INC1392948",
60     "bot": "BOT-Notification",
61     "description": "Failed to send the system alert"
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65 # =====
66 # 1. ADD A NEW INCIDENT TICKET
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```
6. Exit
=====
Enter your choice: 1

--- ADD NEW INCIDENT TICKET ---
Enter Incident ID: INC1392949
Enter Bot: Notification
Enter Short Description: Failed to send the system update

Incident ticket added successfully!
```

Displaying Tickets



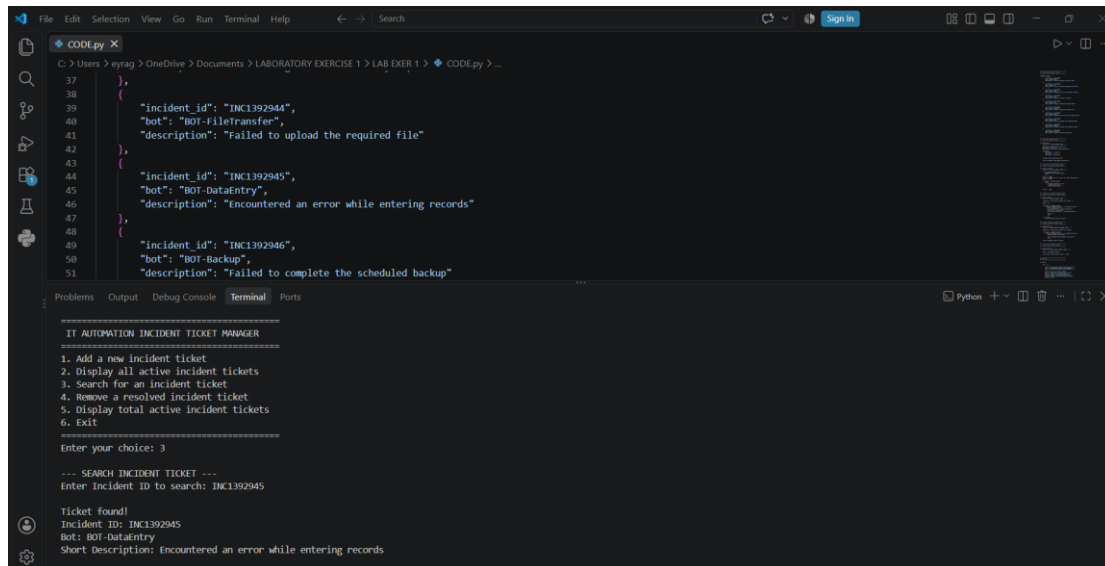
The screenshot shows the same code editor as before, but the terminal output now displays a list of active incident tickets in a table format. The user has entered '2' to view the active tickets.

```
6. Exit
=====
Enter your choice: 2

--- ACTIVE INCIDENT TICKETS ---

Incident ID      Bot              Short Description
-----
INC1392939      BOT-Inventory    Failed to generate the daily report
INC1392940      BOT-Email        Failed to send the scheduled notification
INC1392941      BOT-DataSync     Encountered an error during data transfer
INC1392942      BOT-Invoice      Failed to process an invoice
INC1392943      BOT-Report       Failed to generate the weekly report
INC1392944      BOT-FileTransfer Failed to upload the required file
INC1392945      BOT-DataEntry    Encountered an error while entering records
INC1392946      BOT-Backup       Failed to complete the scheduled backup
INC1392947      BOT-Validation   Failed to validate the submitted records
INC1392948      BOT-Notification Failed to send the system alert
INC1392949      notification      Failed to send the system update
```

Searching for a Ticket



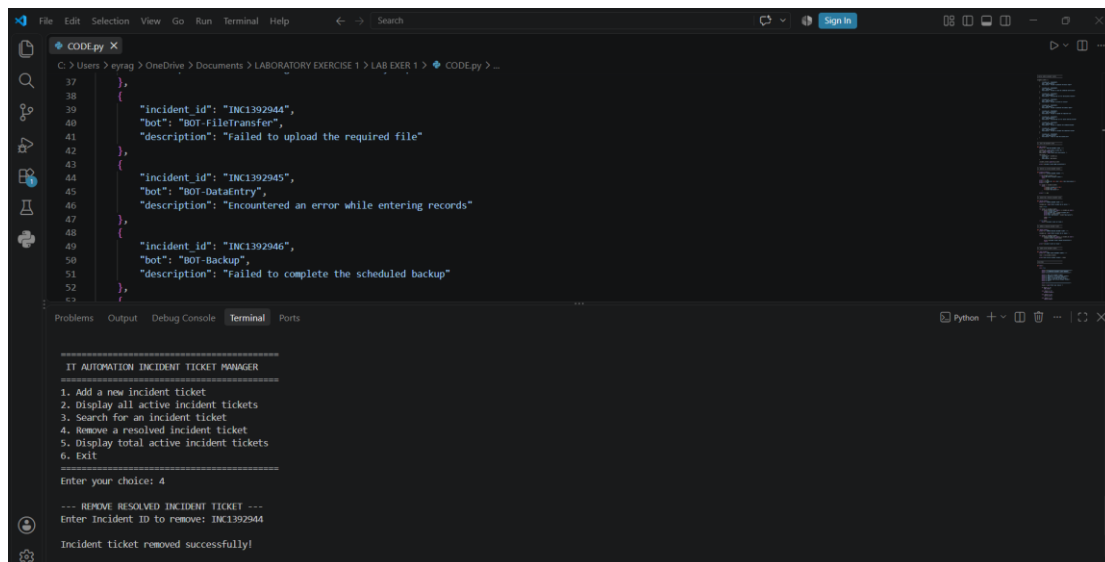
The screenshot shows a Code Editor window with a Python script and its terminal output. The script is a menu-driven program for an IT Automation Incident Ticket Manager. The terminal output shows the user selecting option 3 to search for a ticket, entering the incident ID 'INC1392945', and receiving the ticket details.

```
C:\Users\eyrag> OneDrive\Documents\LABORATORY EXERCISE 1> LAB EXER 1> CODE.py ...  
37  
38  
39     "incident_id": "INC1392944",  
40     "bot": "BOT-Filetransfer",  
41     "description": "Failed to upload the required file"  
42 },  
43  
44     "incident_id": "INC1392945",  
45     "bot": "BOT-DataEntry",  
46     "description": "Encountered an error while entering records"  
47 },  
48  
49     "incident_id": "INC1392946",  
50     "bot": "BOT-Backup",  
51     "description": "Failed to complete the scheduled backup"  
...  
IT AUTOMATION INCIDENT TICKET MANAGER  
=====  
1. Add a new incident ticket  
2. Display all active incident tickets  
3. Search for an incident ticket  
4. Remove a resolved incident ticket  
5. Display total active incident tickets  
6. Exit  
=====
```

Enter your choice: 3
--- SEARCH INCIDENT TICKET ---
Enter Incident ID to search: INC1392945

Ticket found!
Incident ID: INC1392945
Bot: BOT-DataEntry
Short Description: Encountered an error while entering records

Removing a Ticket



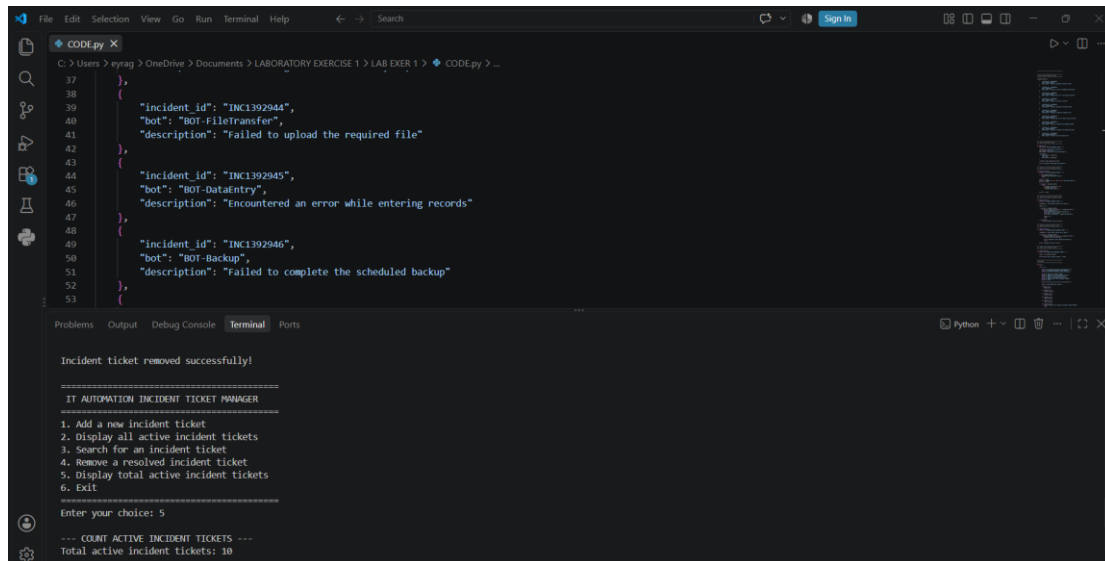
The screenshot shows the same Code Editor window as before, but the terminal output now shows the user selecting option 4 to remove a resolved incident ticket. The user enters the incident ID 'INC1392944', and the program confirms that the ticket has been removed successfully.

```
IT AUTOMATION INCIDENT TICKET MANAGER  
=====  
1. Add a new incident ticket  
2. Display all active incident tickets  
3. Search for an incident ticket  
4. Remove a resolved incident ticket  
5. Display total active incident tickets  
6. Exit  
=====
```

Enter your choice: 4
--- REMOVE RESOLVED INCIDENT TICKET ---
Enter Incident ID to remove: INC1392944

Incident ticket removed successfully!

Counting Tickets



The screenshot shows a code editor with a Python script for an incident ticket manager. The script defines a list of incident tickets and implements functions to add, display, search, remove, and count them. The terminal output shows the program running and displaying the list of tickets.

```
37 },
38 {
39     "incident_id": "INC1392944",
40     "bot": "BOT-Filetransfer",
41     "description": "Failed to upload the required file"
42 },
43 {
44     "incident_id": "INC1392945",
45     "bot": "BOT-DataEntry",
46     "description": "Encountered an error while entering records"
47 },
48 {
49     "incident_id": "INC1392946",
50     "bot": "BOT-Backup",
51     "description": "Failed to complete the scheduled backup"
52 },
53 }
```

Incident ticket removed successfully!

IT AUTOMATION INCIDENT TICKET MANAGER

1. Add a new incident ticket
2. Display all active incident tickets
3. Search for an incident ticket
4. Remove a resolved incident ticket
5. Display total active incident tickets
6. Exit

Enter your choice: 5

--- COUNT ACTIVE INCIDENT TICKETS ---
Total active incident tickets: 10

Conclusion

Conclusion

The laboratory activity demonstrated how a linear data structure can be used to manage incident tickets in an IT automation scenario. The Python list was used to store the incident tickets, while different operations were implemented to add, display, search, remove, and count tickets. Through this activity, the basic concepts and operations of linear data structures were applied to a practical IT-related problem.